Zecpath AI Scalability Strategy

Objective

Design a scalable architecture capable of supporting enterprise-level recruitment.

Horizontal Scaling

Client Requests

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Load Balancer (NGINX / AWS ELB)

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AI Service Cluster

- Resume Parser Service
- ATS Service
- Screening AI
- HR Interview AI
- Technical AI
- Machine Test AI
- Decision AI

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Redis Cache

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PostgreSQL Database

Scaling Strategy

ATS Service

Scaling Level: High

Reason: Large resume processing volume

Screening AI

Scaling Level: Medium

Reason: Moderate interview traffic

Interview AI

Scaling Level: High

Reason: Real-time interview sessions

Decision Engine

Scaling Level: Low

Reason: Small computation

Queue Processing

Heavy Tasks

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Kafka / RabbitMQ

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Worker Nodes

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Database

Caching

- Redis
- ATS Results
- Embeddings
- Frequently accessed reports

Future Improvements

- Kubernetes Auto Scaling
- GPU Inference Servers
- Distributed AI Models
- Edge Deployment
- Multi-region deployment

Expected Benefits

- High availability
- Fault tolerance
- Faster response times
- Lower infrastructure costs